Multimodal World Models

End-of-Semester Independent Study Report

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1. Overview and Motivation

This independent study investigates the design, training and evaluation of a multimodal world model for autonomous driving, grounded in the nuScenes dataset. The core goal is to build a system capable of predicting future sensor observations spanning RGB cameras and LiDAR point clouds, while simultaneously regressing control actions (steering, throttle and brake) across multiple temporal horizons.

The work extends the MUVO (Multimodal World Model with Geometric Voxel Representations) architecture, adapting it to nuScenes.

This report documents progress made during the semester, covering the problem setup, implemented solutions, observed experimental results and an interpretation of what has and has not yet been achieved.

2. Problem Statement and Key Challenges

2.1 Problem Formulation

The task is to train a recurrent probabilistic world model that, given a history of multi-sensor observations and actions, predicts future observations at horizons.

The model is grounded in a GRU-based recurrent dynamics core that maintains a deterministic hidden state capturing temporal context, on top of which posterior and prior distributions over stochastic latent variables are modelled. Decoders branch from sampled latent states to produce each output modality.

2.2 Core Challenges

- **Multi-task, multi-horizon objective conflict:** Six distinct loss types (spatial regression, KL, segmentation, SSIM, action, chamfer) operate at different scales and with different gradient magnitudes. Naively summing them creates competition between tasks and risk of one dominating training.

- **Class imbalance in LiDAR segmentation:** BEV segmentation is heavily dominated by the background class, which routinely achieves an IoU near 0.987. This saturation makes the metric an unreliable progress signal and may encourage the model to ignore rare foreground classes. The issue also stems from the fact that BEV segmentation here is termed as a binary class classification problem of whether an object is present in the space or not.
- **Training instability in later steps:** Segmentation, probabilistic (KL), and RGB losses all exhibit large, sudden spikes in the second half of training, suggesting gradient explosions, learning rate schedule issues, or negative task interference as tasks compete for shared representations.
- **Computation times and OOM issues:** In terms of computation overhead, time per epoch and OOM issues due to RAM allocation blowing up.
- **Temporal horizon decay:** Predicting further into the future compounds uncertainty; horizon-4 predictions should naturally incur higher loss, yet the multi-task weighting must remain coherent across all horizons simultaneously.
- **Actions Integration:** Adapting to nuScenes dataset also meant choosing and injecting action sequences in a temporally cohesive manner so as to match the respective scene data. Misalignment at this stage would silently corrupt the control regression task and the conditioning of future predictions, making it one of the more subtle but consequential integration challenges.
- **Bottlenecks:** In transformer block during tokenization and transition phase of world model.

3. Architecture and Implemented Approach

3.1 Model Architecture

The implemented model follows a four-stage pipeline:

- **Modality-specific encoders:** RGB images are processed through a ResNet backbone. LiDAR point clouds are encoded through a dedicated spatial encoder. Each encoder produces a fixed-dimensional latent embedding.
- **Cross-modal fusion:** Transformer-based cross-attention fuses the per-modality embeddings, allowing the model to attend across sensor streams before forming a joint representation. The transformer block was simplified so as to not overfit and overcomplicate the nuScenes adaptation from the intended CARLA dataset. Additionally, metrics in the latent space were calculated to monitor the quality of embeddings, before and after they interact with each other through the transformer.
- **Recurrent world model core:** A Gated Recurrent Unit (GRU) maintains a deterministic hidden state that accumulates temporal history. This state seeds posterior and prior distributions over stochastic latent variables, enabling both temporal consistency and uncertainty modelling.
- **Multi-modal decoders:** Separate decoder heads reconstruct RGB images, LiDAR 3D point clouds and BEV segmentation maps from sampled latent states. Due to the BEV segmentation loss and computational overhead issue mentioned above, the BEV segmentation map decoding was disabled as it was not helping or affecting the model in training.

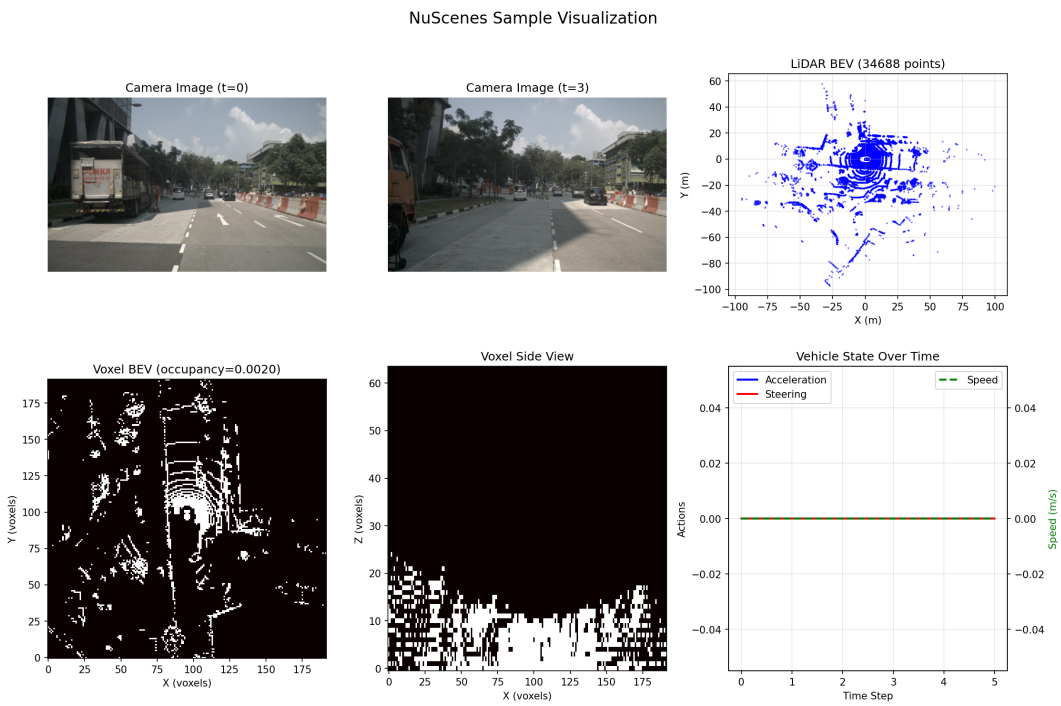


Figure 1 Sample Visualisation with unprocessed(dummy) actions

Substantial engineering work was undertaken to make the pipeline practically trainable within the constraints of the HiPerGator cluster environment. Several targeted modifications were developed and validated, each addressing a specific bottleneck in training throughput, data handling or pipeline correctness.

- **Checkpoint resumption and staged training**: The training loop was extended to support both explicit checkpoint resumption and automatic resume from the most recent saved state. This was essential for running multi-stage jobs on HPG, where wall-time limits require training to be split across multiple job submissions. Without this mechanism, every interrupted run would require restarting from scratch, making longer training runs infeasible.

- **Memory and OOM mitigation:** Out-of-memory (OOM) failures were encountered due to the high memory demands of the transformer cross-attention block and the recurrent world model transition stage. These were addressed through a combination of reduced sequence lengths, gradient checkpointing where applicable, quantization of data and simplification of the transformer architecture (discussed further below). Resolving these bottlenecks was a prerequisite for stable multi-step training runs.

- **Custom dataset fraction support:** A configurable fraction parameter was added to the nuScenes data loader, enabling training on a user-specified subset of the dataset. This allowed quick hypothesis testing and architectural changes to be validated at reduced scale before committing to full-dataset runs. Combined with the LiDAR subsampling described below, this reduced per-epoch training time from approximately 3–4 hours to roughly 30 minutes under current configurations, dramatically accelerating the iteration cycle.

- **LiDAR point cloud subsampling**: Raw nuScenes LiDAR scans contain up to 30,000 points per frame, which created a significant compute and memory bottleneck when fed directly into the encoder. A uniform subsampling strategy was implemented, selecting every n-th point per scan to yield a fixed budget of 2,000 points per frame (configurable via the training config). This approach preserves the

angular structure and spatial distribution of the original scan while delivering consistent tensor shapes across frames and scenes. Temporal consistency across scenes was maintained throughout, ensuring that subsampled sequences remain coherent for the recurrent dynamics model.

- **nuScenes CAN bus integration and validation**: Integrating nuScenes CAN bus telemetry into the pipeline required careful attention to data alignment, normalization and temporal ordering. The integration was validated end-to-end across five criterias: (1) CAN bus data correctly read via the NuScenesCanBus API; (2) steering angle and longitudinal acceleration properly extracted from the raw telemetry; (3) data alignment between scenes and their corresponding action sequences verified; (4) temporal coherence of action sequences confirmed across all scenes and (5) action values properly normalized prior to being fed to the control regression heads. This validation process ensured that the control regression task is grounded in correctly aligned, physically meaningful action signals.

- **Encoder–decoder validation and transformer simplification:** To verify that the RGB and LiDAR encoder–decoder blocks produce quality token representations before and after fusion, each sub-pipeline was tested in isolation and in combination: RGB-only reconstruction, LiDAR-only reconstruction, and joint RGB+LiDAR fused token reconstruction were each evaluated independently. This systematic ablation revealed that the original transformer fusion block was over-parameterized relative to the complexity of the data and the needs of the downstream pipeline, acting as a representational bottleneck rather than an enhancement. The transformer architecture was subsequently simplified, resolving the bottleneck and allowing both modalities to contribute meaningfully to the fused joint representation. Additionally, using a pre-trained decoder block coupled with the pre-trained encoder yielded in better results even with small training runs which can be seen based on the below metrics:
  Training loss reduced to 0.05 at 20th epoch with validation loss mirroring it at 0.05. Validation SSIM and PSNR was at 0.664 and 21.77 dB respectively, showcasing decent baselines for early runs on the encoder-decoder pipeline. This small pipeline also included observing quality of token embeddings in the latent space through L2 loss (2.79) and cosine similarity(-0.044) metrics which helped validating the reconstruction quality and progress of the pipeline.

Collectively, these modifications transformed the pipeline from a research prototype that was impractical to train at scale into a stable, iterative system capable of completing meaningful training runs within HPG job time constraints.

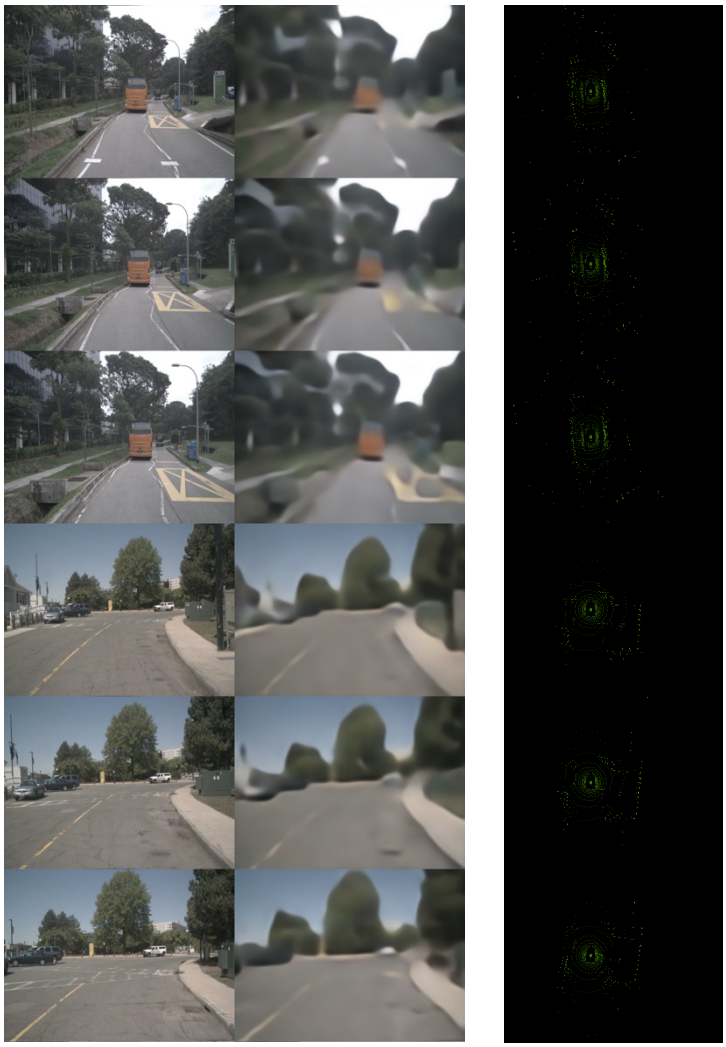


Figure 2 Scene Reconstruction via Encoder-Decoder Pipeline indicating Input, Reconstructed RGB and LiDAR with minimal training

3.2 Training Objectives

The model is trained end-to-end with the following loss components:

Loss Component	Type	Modality	Notes
Spatial Regression (L1/L2)	Reconstruction	RGB, LiDAR Depth	Per-pixel spatial loss
LiDAR Reconstruction	Reconstruction	LiDAR 3D	Evaluated via Chamfer Distance on val
Segmentation Loss (CE)	Classification	BEV / LiDAR Seg.	Cross-entropy; imbalance a concern
KL Divergence	Probabilistic	Latent transitions	Posterior vs prior alignment
SSIM Loss	Perceptual	RGB	1 − SSIM as auxiliary reconstruction signal

Loss Component	Type	Modality	Notes
Action Regression	Control	Steering, Throttle, Brake	Separate regression heads per control signal

3.3 Evaluation Metrics

Validation is tracked across the following metrics for each modality at each checkpoint:

- IoU (background and foreground) - LiDAR BEV segmentation
- Chamfer Distance - LiDAR 3D reconstruction quality
- PSNR and SSIM - RGB reconstruction fidelity
- Validation loss for RGB, LiDAR depth, probabilistic (KL), and segmentation at each horizon
- Action losses (steering, throttle/brake) - control regression accuracy
- Cosine Similarity – reconstruction quality of embeddings and token alignement
- L2/Euclidean Distance – measure for token compactness
- Negative Log Likelihood – calibration for image/lidar segmentation
- Expected Calibration Error – measure for prediction confidence

4. Experimental Results

The last training phase was run for approximately 9,174 gradient steps, with validation checkpointed 22 times across the run. Results are reported below per modality.

4.1 LiDAR Depth Prediction

Training depth loss (horizon 1) began at 0.0808 and steadily decreased to 0.0318 by the end of training, with a minimum recorded value of 0.0313. Validation depth loss tracked this behaviour well, stabilising in the range 0.038–0.058, beginning at 0.044 and ending at 0.038.

Metric	Start (Step 399)	Best	End (Step 8799)
Train Depth Loss (h1)	0.0808	0.0313	0.0318
Val Depth Loss (h1)	0.0436	0.0382	0.0382

4.2 LiDAR 3D Reconstruction (Chamfer Distance)

LiDAR reconstruction loss (training) dropped from 0.0676 to 0.0109 over the course of training, reaching a minimum of 0.0099; a strong absolute improvement. However, validation Chamfer Distance tells a markedly different story: it increased monotonically from 1.71 at step 399 to a peak of 39.92 at step 6399, before partially recovering to 16.91 at the end of the run.

This divergence between training reconstruction loss and validation Chamfer Distance is the most notable anomaly in the results. Possible explanations include distribution mismatch between training and validation

splits, overfitting of the LiDAR reconstruction decoder, or a unit/scale discrepancy between the training loss and the Chamfer Distance metric computation.

Step	Val Chamfer Distance
399	1.71
1,999	6.04
3,999	25.45
6,399	39.92 (peak)
7,599	7.69 (recovery)
8,799	16.91

4.3 LiDAR Segmentation

Training segmentation loss began at 0.0455 and generally trended downward, reaching a minimum of 0.0115. However, it exhibits large late-stage spikes, reaching up to 0.508, roughly 10x the minimum near the end of training. This pattern is consistent across horizons.

Validation background IoU remained very high throughout (≈ 0.987), consistent with the known class imbalance in BEV segmentation where the background dominates.

4.4 RGB Reconstruction

Training RGB loss at horizon 1 started high at 0.877 on validation (first checkpoint) and improved substantially to ≈ 0.274–0.308 in the mid-training region. However, large oscillations emerged later in training, with validation RGB loss spiking to 0.654 (step 6799) before recovering. The final checkpoint sits at 0.298, comparable to the best observed mid-run value.

Training SSIM loss (1 − SSIM) was relatively flat, ranging from 0.026 to 0.060 across the run. This is consistent with the model learning basic spatial structure early but struggling to improve fine perceptual quality thereafter.



Figure 3 Input Image

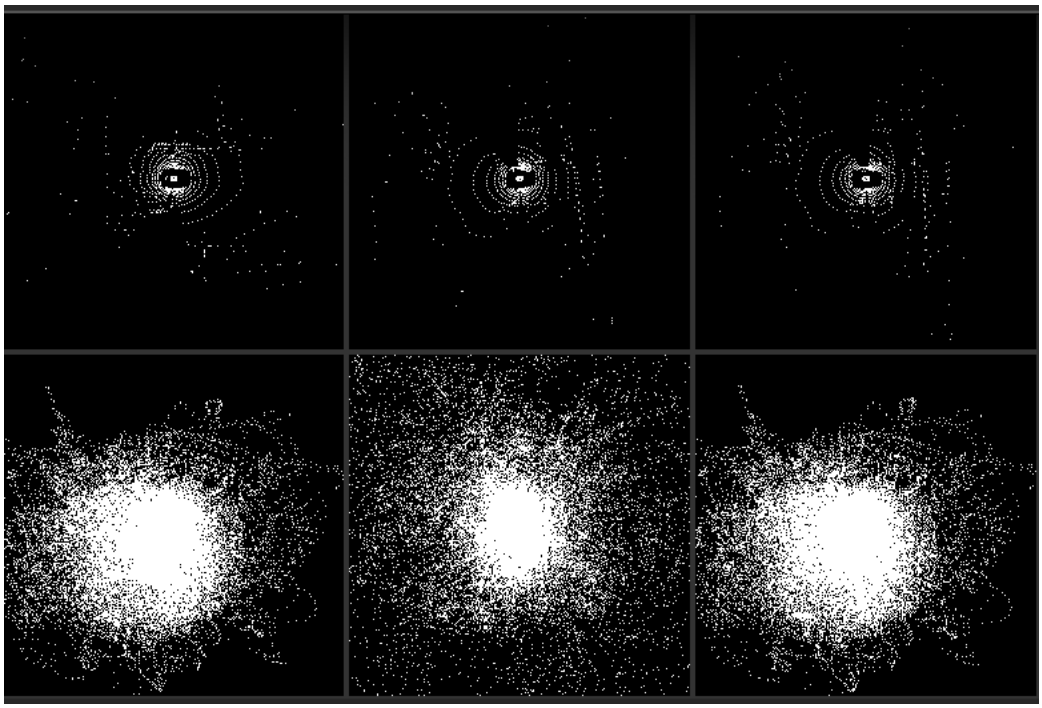


Figure 4 LiDAR Reconstruction(above) and Point Cloud Data of Input Image(below)

4.5 Probabilistic Latent Loss (KL Divergence)

The KL divergence loss between posterior and prior began very low (≈ 0.0038) and remained stable for much of training. Late in the run, large spikes appeared, reaching 0.489 during training. Validation probabilistic loss was similarly stable at approximately 0.007–0.025 through step 6399, then rose to 0.096 at step 7199 and settled around 0.063–0.069 toward the end of the run.

The late-stage KL spikes suggest that the prior and posterior distributions begin to diverge significantly in later training, possibly because the deterministic hidden state has shifted in a way that the prior can no longer track. This may be symptomatic of the broader instability pattern seen across multiple loss components.

4.6 Control Regression: Steering and Throttle/Brake

Steering loss began high (≈ 0.543 at step 24) and decreased to 0.112 by the end of training, though with high variance throughout (min: 0.014, max: 0.586). This reflects the inherent difficulty of steering prediction: it is continuous, sensitive to perception quality and prone to data distribution imbalance across scene types.

Throttle/brake loss remained very small throughout training, ranging from **0.000462 to 0.015**. This is consistent with throttle and brake being more constrained and predictable signals than steering in typical driving datasets, where the vehicle is often at a near-constant speed or braking in well-defined conditions.

Control Signal	Loss at Step 24	Loss at Step 9174	Min	Max
Steering	0.5426	0.1116	0.0140	0.5857
Throttle / Brake	0.0111	0.0039	0.000462	0.0152

5. Interpretation and Analysis

5.1 What Has Been Achieved

- **Stable LiDAR depth prediction:** The validation depth loss converged cleanly, indicating the model has learned a reliable geometric depth signal. This is the most confidently positive result from the experiment.
- **RGB reconstruction shows early learning:** The large drop in validation RGB loss from 0.878 to ~0.298 demonstrates that the model does learn meaningful image structure. The subsequent instability, rather than reflecting catastrophic forgetting, more likely reflects the tension between the RGB objective and competing tasks in the multi-task setting.
- **Control regression partially successful:** Steering loss showed improvement over training. Throttle/brake was solved quickly, suggesting the action regression heads are functioning correctly and that the learned latent state contains some behaviourally relevant information.
- **Probabilistic modelling is functional:** KL loss values remained low and stable for the majority of training, suggesting the posterior-prior alignment is working. The late spikes are a stability concern but do not negate the core mechanism.
- **Additional Observed Metrics:** Prototyping of latent space quality monitoring through reconstruction proved successful as cosine similarity and L2 distance provided interpretable signals about embedding quality and token compactness. NLL showed mixed results with notable fluctuations, which is indicative of an incorrect or inconsistent implementation within the current pipeline rather than a fundamental model failure. ECE results confirm that the model is currently underconfident in its predictions, which is expected at this stage given that a strong baseline has not yet been established.

Additionally, uncertainty heatmaps were integrated into the pipeline to spatially localise where the model is least confident in its predictions. Initial results are promising as a diagnostic tool, though outputs varied considerably, attributable in part to insufficient training steps and the difficulty of reliably mapping latent-space uncertainty signals back to pixel-space RGB predictions. This remains an active work in progress.

5.2 Key Concerns and Open Issues

- **Chamfer Distance divergence on validation:** The consistent increase in validation Chamfer Distance despite decreasing training reconstruction loss obviously does not translate well. The model may be memorising training geometry without generalising to validation scenes. Alternatively, a preprocessing or normalisation inconsistency between training and evaluation may be inflating the metric. This requires further investigation.
- **Late-stage training instability:** Spikes appearing simultaneously across segmentation, KL and RGB losses in the second half of training suggest a systemic issue; likely the learning rate remaining too high after a period of stability, or accumulated gradient noise from hard examples. A cosine decay or warmup-and-decay schedule could help based on the observation.
- **Segmentation class imbalance:** The background IoU of 0.987 provides no signal about foreground class learning. Per-class IoU for vehicles, pedestrians and other rare classes is needed to meaningfully evaluate segmentation quality.
- **Interpretability and generalization studies not yet conducted:** The original proposal included attention visualization, latent probing and modality dropout experiments. These ablations remain as future work for the next phase of the study.

6. Next Steps and Planned Work

The following directions are planned for the continuation of this study:

6.1 Immediate Fixes

- **Investigate Chamfer Distance divergence:** Compare normalisation of LiDAR point clouds between training and evaluation splits; verify the Chamfer Distance implementation operates in the same coordinate frame as the training loss.
- **Address training instability:** Introduce a learning rate decay schedule (e.g., cosine annealing) to prevent late-stage loss spikes as the probabilistic branch may not be learning meaningful uncertainty structure. Adding a per task gradient norm monitoring can also be helpful.

6.2 Architecture and Training Improvements

- Experiment with loss weighting to address competing objectives.
- Introduce per-class IoU logging to properly evaluate segmentation quality beyond the background class.
- Train for additional steps with a stabilised schedule and evaluate whether the Chamfer Distance trend reverses.

6.3 Research Objectives (Original Proposal)

- **Interpretability:** Conduct latent probing experiments to assess whether scene attributes (object category, ego-motion) are linearly decodable from the learned latent space. Additionally, attention visualisation over cross-modalities could also be monitored.
- **Generalization/Robustness:** Conduct modality dropout ablations by training and evaluating under RGB-only and LiDAR-only configurations to assess the fusion module's robustness.
- **Comprehensive Evaluation Report:** Produce the multi-dimensional evaluation framework described in the proposal, incorporating calibrated uncertainty metrics, multi-step rollout error curves, and attention entropy analysis.

7. Conclusion

This semester established the training infrastructure, multi-task objective formulation, and baseline experimental results for a multimodal world model built on the MUVO architecture and adapted to the nuScenes dataset. The most concrete achievement is stable LiDAR depth convergence, with meaningful progress in RGB reconstruction and control regression. Multiple reliability issues, particularly the Chamfer Distance divergence on validation must be resolved before stronger quality conclusions can be drawn. Similarly, segmentation and probabilistic branches show instability that must be addressed before meaningful uncertainty quantification experiments can be run.

The core architecture is in place and training is functional. The next phase will focus on stabilising training, verifying metrics, and beginning the interpretability and generalization studies described in the original proposal.